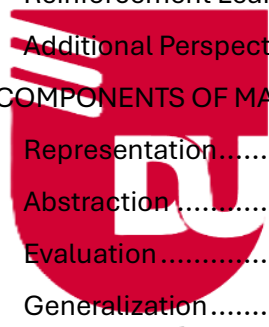


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UNIT 1: ARTIFICIAL INTELLIGENCE(AI)

Based on Capabilities

Narrow AI (Weak AI)

Narrow AI, also known as **Weak AI** or **Artificial Narrow Intelligence (ANI)**, refers to artificial intelligence systems designed to perform a specific task or a set of closely related tasks.

Unlike **General AI**, which aims to replicate human intelligence across a wide range of activities, Narrow AI operates within a limited scope and cannot adapt or generalize its knowledge beyond its programmed domain.

Narrow AI systems are prevalent in today's technology landscape and excel at singular tasks such as facial recognition, language translation, or playing chess. These systems often surpass human performance in their specific domains but lack consciousness, self-awareness, or the ability to think abstractly.

General AI (Strong AI)

Artificial general intelligence (AGI) is a hypothetical type of [artificial intelligence](#) that matches or surpasses human capabilities across virtually all cognitive tasks.^[1]

Beyond AGI, [artificial superintelligence](#) (ASI) would outperform the best human abilities across every domain by a wide margin.^[2] Unlike [artificial narrow intelligence](#) (ANI), whose competence is confined to well-defined tasks, an AGI system can generalise knowledge, transfer skills between domains, and solve novel problems without task-specific reprogramming.

Superintelligence (Super AI)

Artificial superintelligence represents the theoretical pinnacle of machine intelligence—a software-based system that exceeds human cognitive abilities across every domain of knowledge and reasoning. Unlike today's narrow AI systems that excel at specific tasks, ASI would demonstrate superior performance in creativity, problem-solving, social interaction, and scientific discovery.

Current AI systems operate within strict boundaries. Your smartphone's virtual assistant excels at voice recognition but cannot solve complex mathematical theorems. Self-driving cars navigate roads brilliantly yet lack the ability to compose poetry. ASI transcends these limitations, combining and surpassing human-level capabilities across all intellectual domains simultaneously.

Based on Functionalities

Reactive Machines

Reactive Machine AI is the most basic form of artificial intelligence, designed to respond to current inputs without storing past experiences. It operates purely on **present data**, making decisions based on predefined rules or algorithms without any memory or learning capability.

These systems are **task-specific** and deterministic — the same input will always produce the same output. They are ideal for scenarios requiring **fast, consistent responses**.

Core Characteristics

- **No Memory:** Cannot recall past states or learn from history.
- **Rule-Based Decisions:** Operates on fixed logic or behaviour tables.
- **High Speed:** Processes inputs and produces outputs almost instantly.
- **Predictable Behaviour:** Always follows the same decision path for identical inputs.

Limited Memory AI

Limited Memory AI refers to systems that can store and use past data for a short period to improve decision-making. Unlike reactive AI, it leverages recent historical information without retaining it permanently, making it ideal for real-time, context-aware applications.

Key traits include **temporary data storage**, reliance on **machine learning models**, and **real-time adaptability** across industries like transportation, e-commerce, and robotics.

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Theory of Mind AI

Theory of Mind (ToM) in AI refers to the ability of machines to **infer and comprehend the mental states of humans and other agents**, allowing them to anticipate behaviour, intentions, and emotions rather than merely reacting to observable inputs. This capability is inspired by human cognitive development, where understanding that others have beliefs and desires different from one's own is crucial for social interaction. Unlike reactive or limited-memory AI, ToM AI focuses on **social cognition and empathy**, enabling more nuanced interactions.

Self-Aware AI

Self-aware AI is considered the most advanced form of artificial intelligence, theoretically capable of **understanding its own existence, reasoning, and emotions**, in addition to performing tasks and learning from data. Unlike traditional AI, which operates based on pre-programmed rules or learned patterns.

UNIT 2: MACHINE LEARNING(ML)

PARTS OF MACHINE LEARNING

Supervised Learning

Supervised learning involves training algorithms on **labelled data**, where the input-output relationship is known. The model learns to map inputs to outputs and can make predictions on new, unseen data. Common applications include **classification** (e.g., spam detection, medical diagnosis) and **regression** (e.g., predicting house prices or stock values). Popular algorithms include **linear regression, logistic regression, decision trees, support vector machines (SVM), k-nearest neighbours (k-NN), Naive Bayes, and random forests**

Unsupervised Learning

Unsupervised learning works with **unlabelled data** to identify patterns, groupings, or structures within the dataset. It is often used for **clustering, dimensionality reduction, and association rule mining**. Examples include **customer segmentation, market basket analysis, and anomaly detection**. Clustering methods can be **centroid-based, distribution-based, connectivity-based, or density-based**, while dimensionality reduction techniques help simplify data while retaining important information.

Reinforcement Learning

Reinforcement learning trains an **agent to make decisions by interacting with an environment** and receiving feedback in the form of rewards or penalties. The goal is to learn a strategy or policy that maximizes cumulative rewards over time. Methods include **model-based and model-free approaches**, with further distinctions such as **value-based and policy-based methods**. Applications include **robotics, game playing, and autonomous systems**

Additional Perspectives

Some sources also describe machine learning in terms of **components of the learning process**, including **data storage, abstraction, generalization, and evaluation**, which highlight the workflow and reasoning behind how models learn from data. While these components are not divisions of ML per se, they provide insight into the stages involved in building and refining machine learning models.

In summary, **the three primary divisions—supervised, unsupervised, and reinforcement learning—cover the majority of machine learning approaches**, each suited to different types of data and problem-solving scenarios.

COMPONENTS OF MACHINE LEARNING

Representation

This refers to the way knowledge is represented for ML purposes. Some examples include decision trees, sets of rules, instances, graphical models, neural networks, support vector machines, model ensembles, and various others.

Abstraction

Abstraction simplifies the representation of a problem, allowing for more efficient problem-solving with reduced memory and computation requirements. Examples of data abstraction are decreasing the spatial and temporal resolution or dividing continuous variables into meaningful ranges that align with specific goals.

Evaluation

Every ML project needs a method for evaluating hypotheses. Some examples are accuracy, prediction and recall, squared error, KL divergence (relative entropy), and others.

Generalization

Generalization is crucial for a model to effectively handle new, unfamiliar data that comes from the same distribution as the data used to train the model. It allows teams to gain a deeper understanding of overfitting and assess the quality of a model.

Data Storage

This one might easily get forgotten among the components of machine learning, but where your data resides is very important. Common storage solutions for machine learning include object storage, distributed file systems, and cloud-based storage.

UNIT 3: PROBABILITY AND STATISTICS

PROBABILITY

Classical Probability

Classical probability is the likelihood of an event occurring when all possible outcomes are equally likely, calculated as the ratio of favorable outcomes to total outcomes.

Definition and Formula

Classical probability, also known as Laplace's probability, assumes that **all outcomes of an experiment are equally likely**. The probability of an event A is calculated using the formula:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

This formula applies when outcomes are **mutually exclusive** (only one can occur at a time) and **exhaustive** (all possibilities are considered)

Empirical Probability

Empirical probability is the probability of an event estimated from observed data or experimental results rather than theoretical calculations.

Definition and Formula

Empirical probability, also called experimental probability, is calculated based on actual observations or experiments. It is defined as the ratio of the number of times an event occurs to the total number of trials conducted:

Empirical Probability (P(E)) = Number of times event occurs / Total number of trials

For example, if a die is rolled 100 times and the number 6 appears 18 times, the empirical probability of rolling a 6 is $18/100 = 0.18$.

Subjective Probability

Subjective probability is the likelihood of an event occurring based on personal judgment, experience, or belief rather than objective data or calculations.

Definition and Key Features

Subjective probability is a type of probability that reflects an individual's personal belief or opinion about the likelihood of an event. Unlike classical or empirical probability, it **does not rely on historical data, mathematical formulas, or statistical calculations**; instead, it is based on intuition, experience, or expert judgment.

STATISTICS

Descriptive Statistics

Descriptive statistics **summarize and organize data** to make it understandable. It does not make predictions but provides a clear picture of the dataset. Key components include:

- **Measures of Central Tendency:** Indicate the “centre” of data.
 - **Mean:** Average value.
 - **Median:** Middle value in an ordered dataset.
 - **Mode:** Most frequently occurring value.
- **Measures of Variability:** Show how spread out the data is.
 - **Range:** Difference between highest and lowest values.
 - **Variance:** Average squared deviation from the mean.
 - **Standard Deviation:** Typical distance of values from the mean.



Inferential Statistics

Inferential statistics **allow conclusions or predictions about a population** based on a sample. It uses probability theory to make inferences, test hypotheses, and estimate population parameters. For example, analyzing a sample of student test scores to predict overall class performance or comparing results across different years.

Parametric Statistics

Parametric methods assume that the data follows a **specific distribution**, usually normal, and are often applied to continuous data. Examples include t-tests and ANOVA, which rely on assumptions about population parameters.

Non-Parametric Statistics

Non-parametric methods **do not assume a specific distribution** and are suitable for ordinal or categorical data. Examples include the Chi-square test and Mann-Whitney U test, which are used when parametric assumptions are not met.